

Structural Integrity & Health Monitoring – Experimental Coursework

Part 1 – Instructions and Background

This experimental coursework is designed to supplement the knowledge presented in lectures. As such, it will require the synthesis of several topics presented during the Structural Health Monitoring course and require students to apply them to real world measurements obtained in the laboratory.

In completing this coursework, you should create *one* MATLAB or Python script which performs the necessary calculations to answer/plot the questions below. This script should be attached with this document in a *.ZIP* file when submitted.

As this may be your first experience with Structural Health Monitoring experimental equipment, an “*Experimental Coursework Guide*” has been created to assist with data recording during the lab session.

Name:	Zhe Wei Kho	Plate #: PZT Configuration:	Plate 3 Location 1 and 4
CID:	02324081	Chosen Frequency:	50 kHz and 200 kHz

Part 2 – Signal Processing (15%)

2.1) Briefly describe the nature of environmental noise captured in the recordings and whether these hinder the identification of wave packets? What are some possible sources of noise? – 50 Words (5%)

Noise captured in the recordings could have arisen from environmental and technological reasons, such as inter-channel cross-talk (1). While more apparent for the 50kHz frequency, noise did not prevent identification of wave packets after averaging and filtering. Likely sources include accuracy of the signal analyser, cable coupling, and electromagnetic interference. This can be minimised by modifying the transducers (2).

2.2) Select a noise filtering algorithm for use (you do not need to create your own algorithm). Explain why you chose this algorithm, and briefly describe how it works – 50 Words (5%)

Ten readings were taken and the signal was averaged. A 4th-order Butterworth band-pass filter was chosen according to best practices (1,3). This was set to 20-300 kHz for 50 kHz and 100-1000 kHz for the 200 kHz system. The maximally flat response of the filter should attenuate high-frequency noise while preserving dominant waveform characteristics, wave-packet timing, and the signal amplitude.

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2.3) Present a plot showing the frequency content of both the filtered and unfiltered signal from one actuator-sensor path. Explain whether your chosen algorithm has been effective at removing environmental noise from the measurements and why – 50 words (5%)

The algorithm chosen was effective. Comparing signals (a, c) and (b, d) from Figure 2-1, the high-frequency noise superimposed on the waveforms is visibly reduced, as highlighted by the inset plot. The wave packet shape and amplitude are well preserved, which was verified by plotting the envelope using the Hilbert transform.

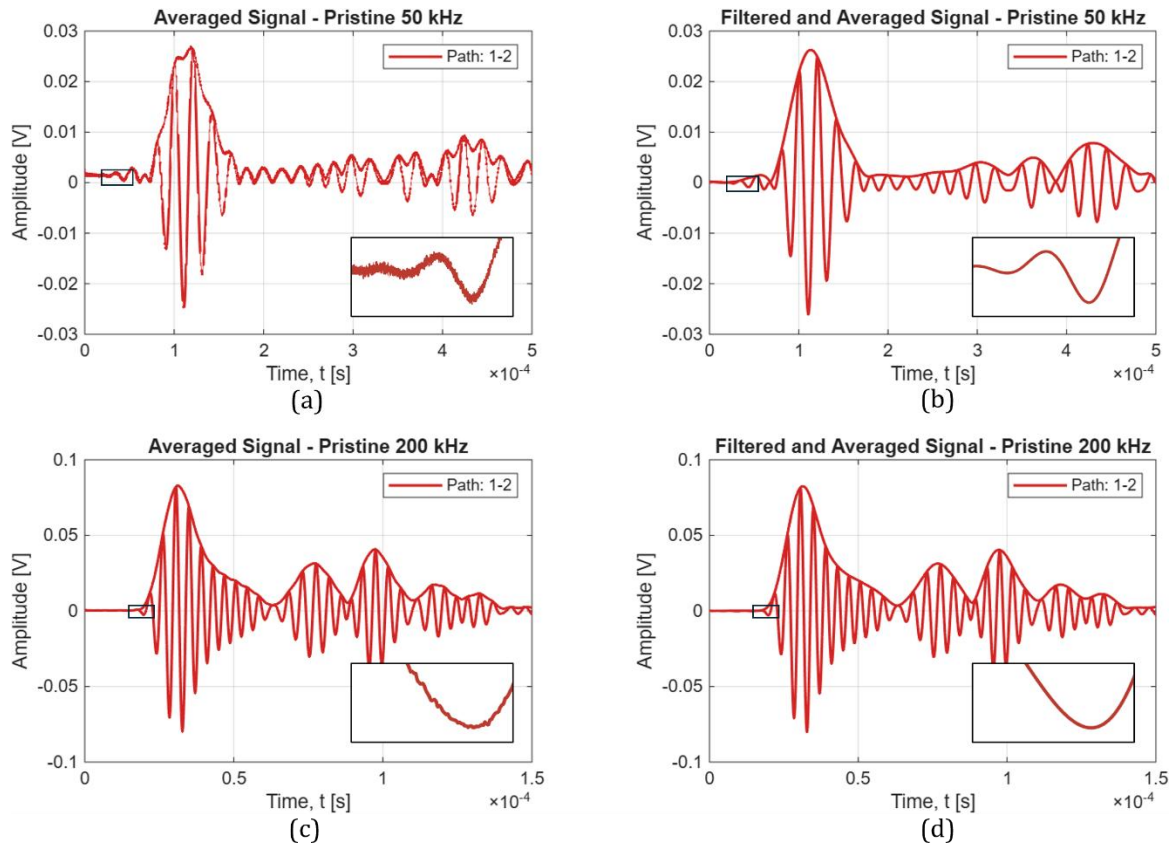


Figure 2-1. Time-domain signals from Path 1-2 of Plate 3 in pristine condition, inset shows zoomed plot. (a) Unfiltered 50 kHz; (b) Filtered 50 kHz; (c) Unfiltered 200 kHz; (d) Filtered 200 kHz.

Part 3 – Wave Modes/Wave Energy (25%)

3.1) Explain the difference in motion between the anti-symmetric and symmetric wave modes. You can create a diagram to help illustrate – 50 Words (5%)

Anti-symmetric (A_0) Lamb wave modes exhibit flexural/radial behaviour whereas symmetric (S_0) modes exhibit axial behaviour (4,5). At a given frequency, A_0 has shorter wavelength, and easier activation of out-of-plane particles resulting in bending motion. In contrast, S_0 tends to travel further, holds higher velocities with predominantly in-plane motion (longitudinally).

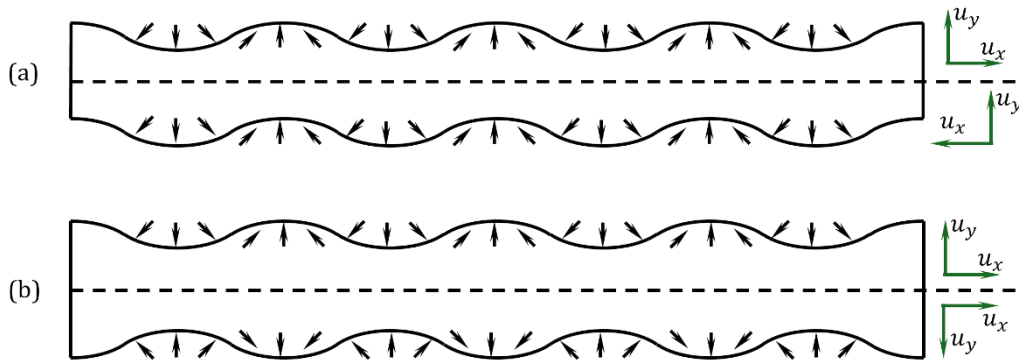


Figure 3-1. Plate under Lamb wave excitation with wave propagating left to right. Arrows indicate particle motion. (a) Anti-symmetric wave mode; (b) symmetric wave mode.

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3.2) Show a plot of one of your recorded actuator-sensor signals (i.e. the voltage versus time on the x-axis). On the plot, show the presence of cross-talk (if any), the anti-symmetric wave packet and the symmetric wave packet (if they exist depending on your excitation frequency). Explain the rationale behind your annotations. If a feature does not exist, explain why you are convinced this is the case – *100 Words (10%)*

The thin carbon-fibre plate means only two modes are expected. The first packet arriving at $t \approx 0.35 \times 10^{-4}$ s (50 kHz) and $t \approx 0.20 \times 10^{-4}$ s (200 kHz) is the S_0 mode as we expect higher group velocity. The A_0 mode is the slower packet arriving at $t \approx 0.75 \times 10^{-4}$ s (50 kHz) and $t \approx 0.65 \times 10^{-4}$ s (200 kHz). In addition, since the A_0 mode is more dispersive at low frequencies, it should have a higher wavelength than S_0 mode at low frequencies. Likewise, S_0 mode is more dispersive at high frequencies, hence it should be wider than other signals.

Cross-talk was seen to be insignificant as no unusual patterns could be identified. The lower amplitude waves that follow likely arise from dispersion or mode conversion at free boundaries (6,7).

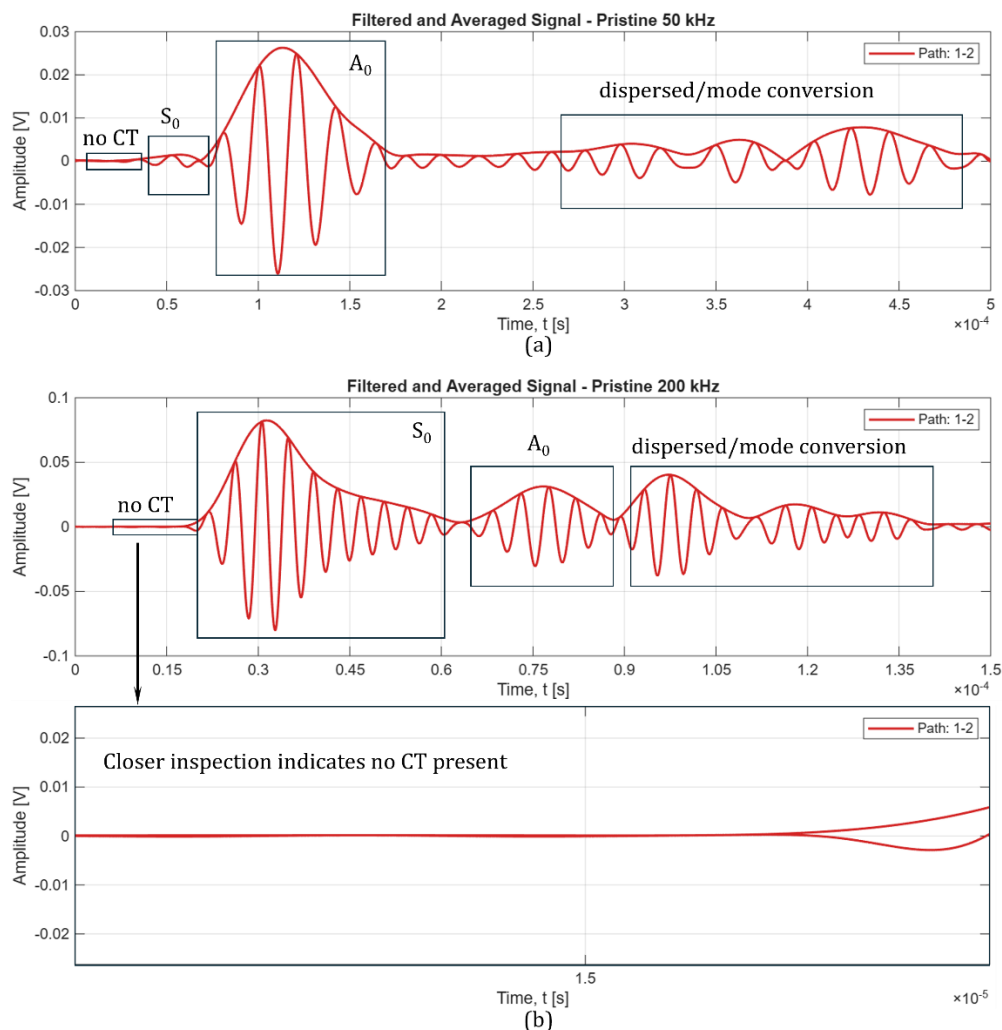


Figure 3-2. Time-domain signal with boxes indicating cross-talk (CT), symmetric wave mode (S_0), anti-symmetric wave mode (A_0), and other waves identified as scattered/dispersed. (a) Filtered 50 kHz; (b) Filtered 200 kHz including close up on CT section.

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3.4) Pick the signal from two or more transducer pairs to show the attenuation of the wave. Is the attenuation frequency dependent? – 50 Words (10%)

Attenuation of A_0 and S_0 modes are frequency dependent. For instance, rate of change in amplitude of S_0 does not decrease at 50 kHz whereas significant attenuation is seen in 200 kHz. This matches expectations as attenuation of lamb waves is described by $u(x, t) = A_0 e^{-\alpha x} e^{i(kx - 2\pi f t)}$, whereas the higher the frequency, f the stronger the attenuation.

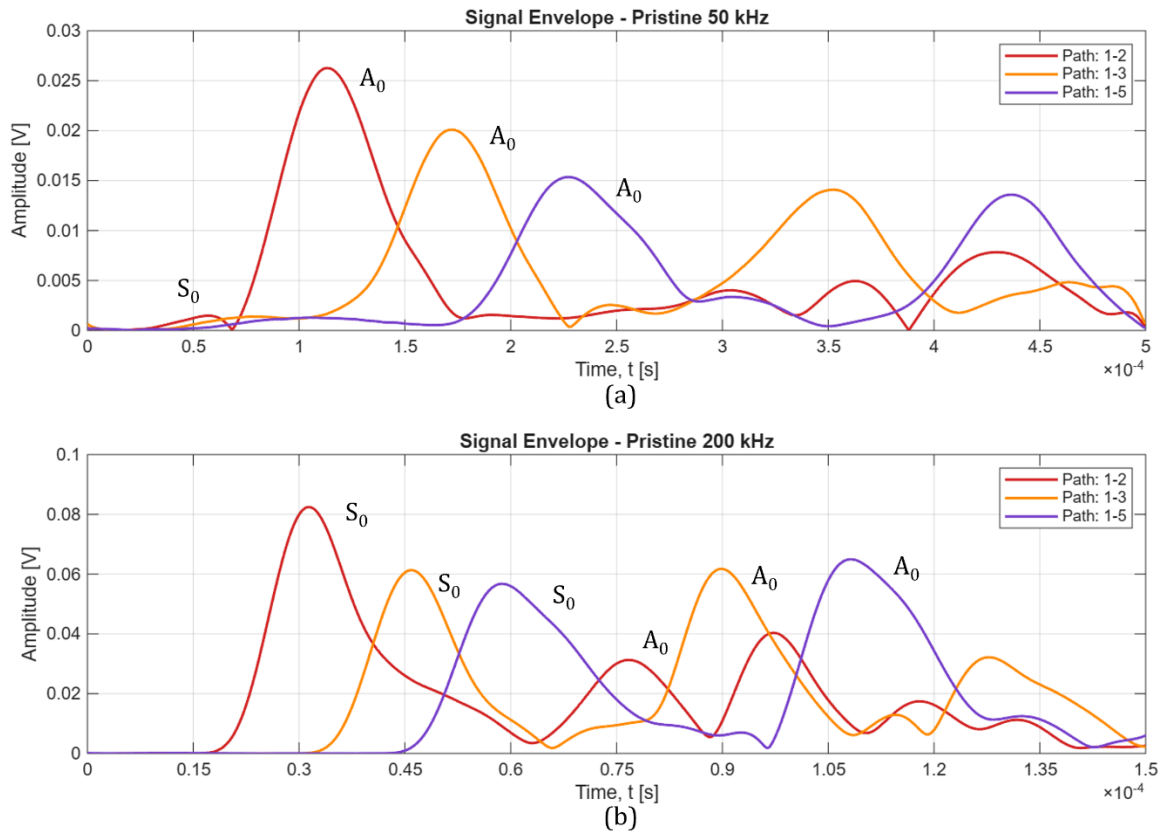


Figure 3-3. Signal envelope of Path 1-2, 1-3, and 1-5.
(a) Filtered 50 kHz; (b) Filtered 200 kHz.

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Part 4 – Group Velocity (20%)

4.1) Provide a plot of the measured sensor signal in a chosen actuator-sensor path. Annotate the point which was used to determine the time of arrival of the first wave packet when calculating the group velocity. Describe how this point was determined algorithmically in your script (It is not acceptable to manually determine the time of arrival of each wave in your script) – 150 Words (5%)

To determine the group velocity, a gated envelope threshold approach was used to calculate the time of arrival (ToA). First, using the Hilbert transform, signal envelope from each set of transducer pairs were plotted. Next, time gates were then established (gate1, gate2) to isolate the wavelet of interest. This allowed maximum for A_0 (50 kHz) and S_0 (200 kHz) to be found without interference from other peaks. Through experimentation, a 10% threshold compatible for all paths was then defined, allowing the ToA of the wave front to be found. As described by Zahra and Aliabadi, this approach avoids any possible interference of cross-talk which can affect the identification of the ToA for each path (1).

To develop the polar representation, eight paths that passed through the center with approximately equal distance were selected. Group velocity was calculated using the *hypot* function for the distance and *atan2d* function for the angle using each actuator-sensor path. This is in line with the approach utilised by Perfetto et al (8).

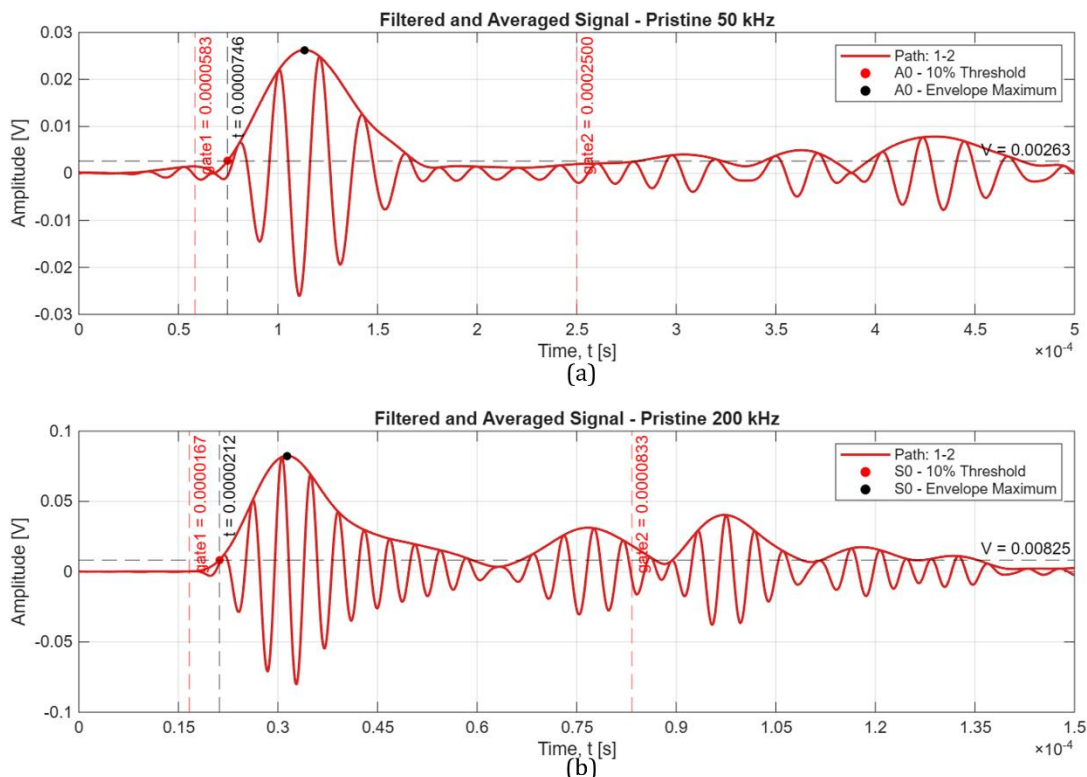
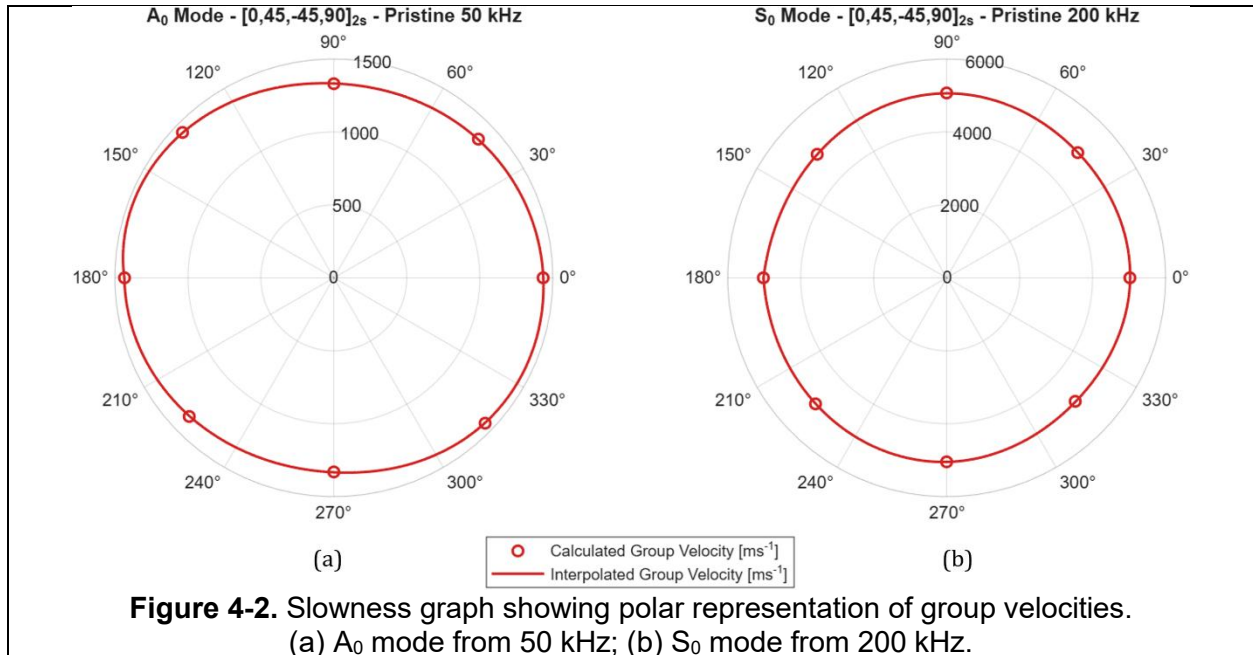


Figure 4-1. Time-domain signal with 10 percent threshold (ToA) and maximum indicated.
(a) Filtered 50 kHz; (b) Filtered 200 kHz.

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4.2) Based on your sensor configuration, indicate, with a polar plot, direction dependent group velocities that you have measured for your plate (there will be a fixed number of angles). Interpolate between the measured angles to provide a continuous radial plot indicating the group velocity of your chosen excitation frequency with respect to radial direction. *0 Words (5%)*



4.3) What was the average group velocity of the first wave packet for your chosen frequency. Briefly explain if this result was expected or not – *50 Words (5%)*

The average group velocity for A_0 mode (50 kHz) was $1,393.0 (\pm 47.1) \text{ ms}^{-1}$ whereas S_0 mode (200 kHz) was $4,987.5 (\pm 61.1) \text{ ms}^{-1}$. The values observed by Perfetto et al were similar with A_0 (50 kHz) at $\sim 1400 \text{ ms}^{-1}$ and S_0 (200 kHz) at $\sim 4500 \text{ ms}^{-1}$. The discrepancies are due to layup sequence anisotropy but also temperature and composite material since [+45/-45/0/90/0]s using M21/T800S plies were used (8).

4.4) Is the group velocity of the first wave packet isotropic with radial direction? Justify your conclusion and explain whether or not this is an expected result – *100 Words (5%)*

The slowness graph shows an approximately circular profile, indicating that group velocity is largely isotropic across all radial directions, with only minor variation between angles. The standard deviation is also small relative to the group velocity at 3.4% (50 kHz) and 1.2% (200 kHz).

This is consistent for both A_0 and S_0 corresponding to expectations as this is a quasi-isotropic layup [0/+45/-45/90]2s. Being balanced and symmetric equates to equal in-plane stiffness (i.e. $E_{11} = E_{22}$). The elastic moduli and density can result in changes in velocity of the signal as noted by Salamone et al (9).

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Part 5 – Damage Detection (40%)

5.1) Choose a damage detection algorithm that can be used to detect the likelihood of damage on a given actuator-sensor path (Hint: A damage detection algorithm should compare the measured pristine and damaged signals on a given actuator-sensor path and account for the fact that damage likely changes the propagation properties of the signal).

Explain how this algorithm works, including how the algorithm produces a likelihood of damage for each actuator-sensor path on the specimen – *200 Words (20%)*

A damage index (DI) approach was used to characterise and localise damage to extract information by comparing pristine and damaged Lamb waves transmitted. In this work, a modified RAPID framework was used with the signal sum of squared difference (SSSD) damage index instead of the correlation coefficient (CC) approach detailed by Zhao et al (10).

SSSD algorithm used is defined as, $DI_n = \frac{\sum_{t=0}^T [c_n(t) - b_n(t)]^2}{\sum_{t=0}^T b_n(t)^2}$, where $c_n(t)$ and $b_n(t)$ represents the current (damaged) signal and baseline (pristine) signal respectively for path, n . This formulation quantifies deviations between the two signals and inherently accounts for damage-induced changes in amplitude, phase, and dispersion related to wave propagation. The combined error makes it one of the effective damage identification algorithms (11).

Signals at 50 kHz containing the A_0 mode were selected due to its sensitivity to surface damages (unlike S_0) and the relatively large simulated damage area by the Bluntack. DI values were computed for all paths and visualised using a bar chart, where a higher DI value indicated a higher likelihood of damage present along the corresponding path.

To enable spatial interpretation, each path DI was multiplied by an elliptical shape function centred on the actuator-sensor line, controlled by scaling parameter, β with decreasing probability radially. Summation of these weighted DIs produced a probability density map, $P(x, y)$ allowing visualisation of damage likelihood across the specimen.

5.2) For the one of a chosen damage location placed **within** the PZT's area of detectability, indicate what are the most affected sensor paths and consequently approximate a "predicted damage location" – *0 Words (10%)*

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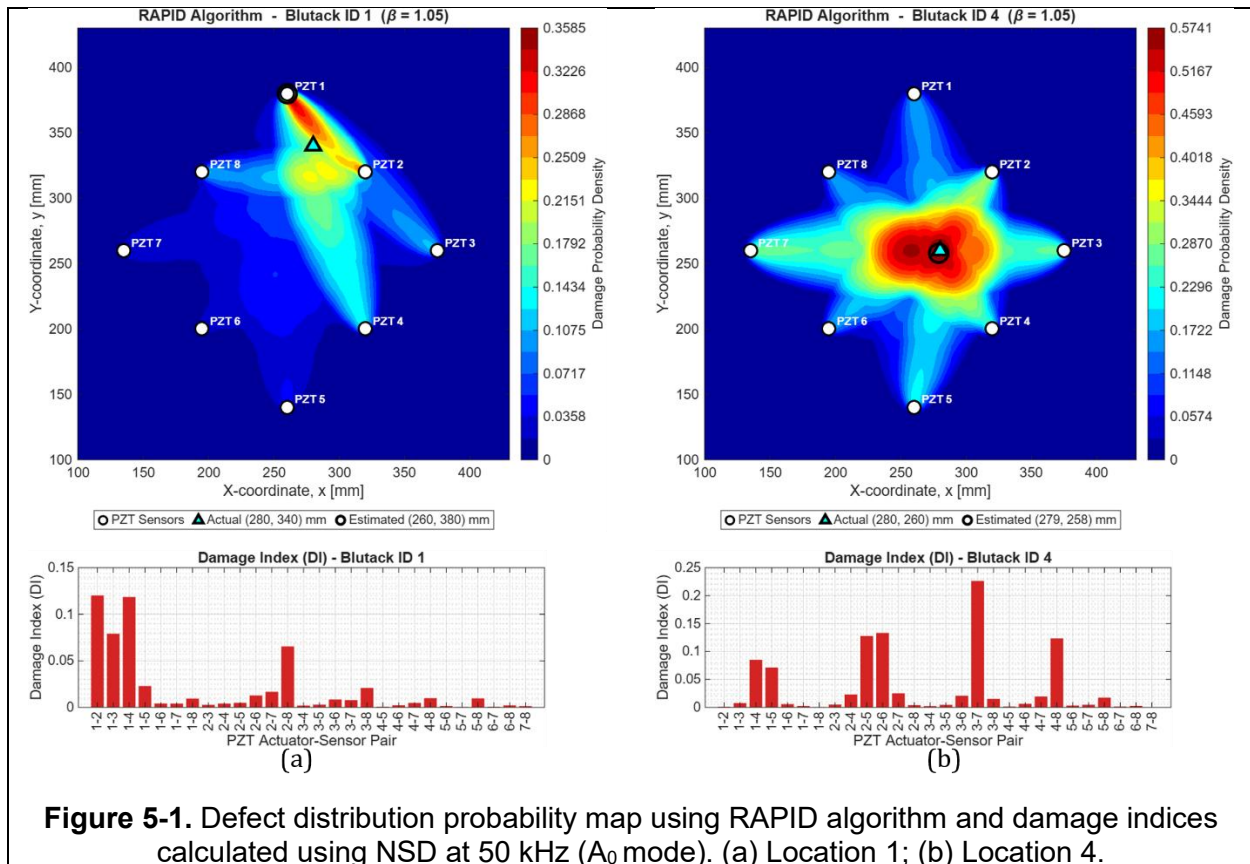


Figure 5-1. Defect distribution probability map using RAPID algorithm and damage indices calculated using NSD at 50 kHz (A_0 mode). (a) Location 1; (b) Location 4.

5.3) For the two damage locations in Question 5.2, briefly discuss whether the chosen algorithm is able to accurately predict the location of damage. List three ways of increasing the accuracy of detection if these panels were to be remanufactured – 200 Words (10%)

The chosen algorithm was unable to accurately estimate the location of damage consistently. While successful for Location 4 ($\Delta \approx 2$ mm), the location was not estimated correctly for Location 1 ($\Delta \approx 45$ mm). This was flagged to be near PZT 1 transducer which is inaccurate and shows the low accuracy with locations on the boundary of the region of interest (ROI).

Reasons this became an issue include a combination of factors such as the algorithm used, transducer layout, and anisotropy. RAPID only provides meaningful imaging at the intersections of the direct paths and works poorly outside the central region. Adopting a different layout could also be considered as the diamond approach here involved overlapping sensors. Moreover, small amount of anisotropy was observed, especially from the 50 kHz slowness graph, which could have affected the accuracy of the plot as an isotropic plate was assumed.

Future improvements could include:

1. Increasing the number of PZT transducers or adopting a more optimised layout to improve coverage (12);
2. Adopting a more sophisticated algorithm such as a hybrid approach with the delay-and-sum algorithm (2);
3. Incorporating correction factors to consider anisotropy using information from slowness/polar graph (8).

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References

References

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Appendix

Data and algorithms used in this assignment can be downloaded through the link below:

https://imperiallondon-my.sharepoint.com/:u:/g/personal/zwk25_ic_ac_uk/IQCPj4E490NSQ6JyL-01-XP4AUBeHGOBQ5LIKshKFQY-TwY?e=TN5mEY